

Estimated-Weight Infinitesimal Jackknife for Moment-Quadratic Estimators

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Purpose

This note fixes the algebra for extending `robust_weighted_moment_ij` beyond complete all-ordinal ULS/DWLS/WLS. The target is the actual estimator whose second-stage weight is built from the data. The companion note `misspec_observed_bread.tex` derives the observed-Hessian bread and the complete all-ordinal diagonal-weight case. Here the goal is the general formula, the full-weight formula, and the implementation grid.

1 Moment estimator and scale

Let $\hat{\kappa}$ be a vector of stage-1 moments with casewise expansion

$$\hat{\kappa} = \kappa_0 + \frac{1}{N} \sum_{i=1}^N g_i + o_p(N^{-1/2}), \quad \frac{1}{N} \sum_i g_i = O_p(N^{-1/2}).$$

For disjoint groups or blocks, stack the block moment vectors and set rows to zero outside the block. The model-implied moments are $\sigma(\theta)$, $d(\theta) = \sigma(\theta) - \hat{\kappa}$, and $\Delta = \partial\sigma/\partial\theta^\top$ is $m \times q$. Equality constraints are represented by a full-rank map K from reduced coordinates α to full θ , and $D = \Delta K$.

The fitted value minimizes

$$F_N(\theta) = \frac{1}{2} d(\theta)^\top \hat{W} d(\theta),$$

or the corresponding block sum. The reduced estimating equation is

$$U_N(\alpha) = D(\alpha)^\top \hat{W} d(\alpha) = 0.$$

At the pseudo-true value, let

$$A = \left. \frac{\partial U(\alpha)}{\partial \alpha^\top} \right|_{\alpha_*}$$

be the observed-Hessian bread. This includes the residual-contracted curvature term from $d(\theta)$; it is not the Gauss-Newton bread except under correct specification.

2 General estimated-weight IJ

Assume the weight has the first-order expansion

$$\hat{W} = W + \frac{1}{N} \sum_{i=1}^N \dot{W}_i + o_p(N^{-1/2}), \quad \sum_i \dot{W}_i = O_p(N^{1/2}),$$

where $\dot{W}_i = \text{IF}_i(\hat{W})$. Perturbing one case gives

$$\delta_i d = -g_i, \quad \delta_i U = D^\top (\dot{W}_i d - W g_i).$$

Define the moment-space row contribution

$$v_i = W g_i - \dot{W}_i d. \tag{1}$$

Then

Theorem 1 (estimated-weight IJ). *For the data-weighted moment-quadratic estimator,*

$$\text{IF}_i(\hat{\alpha}) = A^{-1} D^\top v_i,$$

up to an immaterial global sign convention for g_i . Hence

$$\widehat{\text{Var}}(\hat{\theta}) = K A^{-1} \left\{ \frac{1}{N^2} \sum_{i=1}^N D^\top v_i v_i^\top D \right\} A^{-1} K^\top. \tag{2}$$

Proof. Linearize $0 = U_N(\hat{\alpha})$ around α_* . The data perturbation is $D^\top (\dot{W}_i d - W g_i)$ and the parameter perturbation is $A(\hat{\alpha} - \alpha_*)$. Solving for $\hat{\alpha} - \alpha_*$ yields $A^{-1} D^\top (W g_i - \dot{W}_i d)/N$ per case. Summing the outer products gives (2). $\square$

Corollary 1 (fixed-weight reduction). *If $\dot{W}_i = 0$, $v_i = W g_i$. Since $\hat{\Gamma} = N^{-1} \sum_i g_i g_i^\top$, the IJ covariance in (2) reduces exactly to*

$$K A^{-1} D^\top W \hat{\Gamma} W D A^{-1} K^\top / N.$$

This is the primary regression gate: ULS and any caller-declared fixed weight must be machine-identical to the existing observed-bread sandwich.

3 Weight derivatives

3.1 Diagonal empirical weight

Let $\hat{W} = \text{diag}(\hat{\gamma})^{-1}$ where $\hat{\gamma} = \text{diag}(\hat{\Gamma})$. Then

$$\dot{W}_i = -\text{diag}(\gamma^{-2} \odot \dot{\gamma}_i), \quad v_i = W g_i + \text{diag}(d \oslash \gamma^2) \dot{\gamma}_i.$$

In row form this is the implemented all-ordinal DWLS correction:

$$v_{ik} = g_{ik}/\gamma_k + d_k \dot{\gamma}_{ik}/\gamma_k^2.$$

For complete continuous DWLS, $\dot{\gamma}_i$ is the diagonal of (3). The implementation deliberately discards the off-diagonal empirical-Gamma influence before differentiating $\text{diag}(\hat{\Gamma})^{-1}$; those off-diagonal channels belong to full WLS.

3.2 Full empirical weight

Let $\hat{W} = \hat{\Gamma}^{-1}$. Matrix inverse differentiation gives

$$\dot{W}_i = -W \dot{\Gamma}_i W, \quad v_i = W g_i + W \dot{\Gamma}_i W d.$$

This is the full-WLS target. The required ingredient is the full matrix influence $\dot{\Gamma}_i$, not only its diagonal. A correct implementation must preserve symmetry and include all channels by which the stage-1 sandwich changes.

For complete continuous WLS/ADF, write $x_i = y_i - \bar{y}$ and $c_i = \text{vech}(x_i x_i^\top) - \text{vech}(S_N)$. The moment-influence row is $z_i = c_i$ without meanstructure and $z_i = (x_i^\top, c_i^\top)^\top$ with meanstructure, so

$$\hat{\Gamma} = \frac{1}{N} \sum_{j=1}^N z_j z_j^\top.$$

Under an infinitesimal case-weight perturbation at case i , set $h = x_i$. The exact finite-sample derivative is

$$\dot{\Gamma}_i = z_i z_i^\top - \hat{\Gamma} + \frac{1}{N} \sum_{j=1}^N \left\{ \dot{z}_j(h) z_j^\top + z_j \dot{z}_j(h)^\top \right\}. \quad (3)$$

If mean rows are present, $\dot{z}_{j,\mu}(h) = -h$. For the covariance component indexed by the lower-triangle pair (a, b) ,

$$\dot{z}_{j,ab}(h) = -(h_a x_{jb} + x_{ja} h_b).$$

The derivative of the subtracted covariance S_N is constant across j and therefore drops from (3) because $N^{-1} \sum_j z_j = 0$. The implemented continuous WLS correction uses $d^\top W \dot{\Gamma}_i W$ in row form and is regression-tested against central finite differences of the contaminated empirical weight $\hat{W}_\epsilon = \hat{\Gamma}_\epsilon^{-1}$, including the stacked mean-covariance cross-block.

3.3 DLS and mixed weights

For a Browne-style mixture

$$\hat{W} = \left((1-a)\hat{\Gamma}_0 + a\hat{\Gamma}_1 \right)^{-1},$$

write $M = (1-a)\Gamma_0 + a\Gamma_1$. Then

$$\dot{W}_i = -W \{ (1-a)\dot{\Gamma}_{0i} + a\dot{\Gamma}_{1i} \} W, \quad v_i = W g_i + W \{ (1-a)\dot{\Gamma}_{0i} + a\dot{\Gamma}_{1i} \} W d.$$

If Γ_0 is model-implied and depends only on θ , its derivative belongs in the bread. If Γ_0 is built from saturated sample moments, as in two-stage FIML's normal-theory Stage-2 weight, its casewise derivative is a weight-IJ term. For complete-data continuous DLS with caller-fixed a , $\Gamma_0 = \Gamma_{\text{NT}}(\hat{S})$ and $\Gamma_1 = \hat{\Gamma}_{\text{ADF}}$. The implemented row correction is the same convex mixture of the normal-theory and dense empirical-Gamma IF channels, evaluated with the DLS weight. In meanstructure layouts the normal-theory side is $\text{blockdiag}(\hat{S}, \Gamma_{\text{NT}}(\hat{S}))$ and the empirical side is the full stacked Browne $\hat{\Gamma}_{\text{ADF}}$, including the mean-covariance cross-block. The scalar a is treated as fixed; an empirical-Bayes rule for selecting a would add a separate scalar IF term.

4 Stage-1 sandwich influence

Many magmaan moments are themselves M-estimators. A common form is

$$\hat{\Gamma} = \hat{A}^{-1} \hat{V} \hat{A}^{-\top}, \quad \hat{V} = N^{-1} \sum_i s_i s_i^\top, \quad \hat{A} = N^{-1} \sum_i b_i.$$

The data-direct influence at fixed stage-1 parameter is

$$\dot{\Gamma}_i^{\text{direct}} = A^{-1} (s_i s_i^\top - V) A^{-\top} - A^{-1} (b_i - A) \Gamma - \Gamma (b_i - A)^\top A^{-\top}.$$

If Γ also changes through the movement of the stage-1 parameter κ , add

$$\dot{\Gamma}_i^\kappa = \left[\frac{\partial \Gamma}{\partial \kappa} \right] g_i.$$

The all-ordinal path now computes $\dot{\Gamma}_i^{\text{direct}}$ from integer data and the diagonal of $\dot{\Gamma}_i^\kappa$ by finite differences for DWLS. Full WLS uses the dense matrix $\dot{\Gamma}_i^{\text{direct}} + \dot{\Gamma}_i^\kappa$ with the row correction $d^\top W \dot{\Gamma}_i W$.

5 Implementation grid

Complete all-ordinal. ULS, DWLS, and WLS are implemented for complete integer data. ULS is covered by Corollary 1. DWLS uses the diagonal formula above. Full WLS uses the full $\dot{\Gamma}_i$ matrix in the dense inverse-weight derivative. The dense data-direct channel is gated against case-weight finite differences of the full NACOV matrix; its diagonal extraction is also checked against the DWLS helper. A deterministic resampling-jackknife fixture and misspecification simulation remain useful higher-level checks.

Complete mixed ordinal/polyserial. The same equations apply, but g_i and $\dot{\Gamma}_i$ must cover continuous means, continuous covariances, ordinal thresholds, polychorics, and polyserials in the exact mixed moment order. Fixed-weight ULS is implemented: it only needs casewise g_i and reduces to the observed-bread sandwich. DWLS is also implemented for ordinary complete-data ML/polyserial mixed stats: the diagonal $\dot{\Gamma}_i$ combines the data-direct sandwich channel at fixed mixed kappa with a finite-difference $\partial \text{diag}(\hat{\Gamma})/\partial \kappa$ channel for thresholds, negative means, variances, polychorics, polyserial covariances, and Pearson covariance rows. Full WLS uses the dense mixed $\dot{\Gamma}_i$ with the same data-direct and finite-difference channels, and its diagonal extraction is checked against the DWLS helper.

Complete continuous LS. For ULS with a fixed identity weight, the IJ is only a wrapper around the observed-bread sandwich using raw moment influence rows. This slice has landed as the fixed-weight continuous-LS IJ helper. The sample-built GLS normal-theory-weight slice has also landed: this is a moment-quadratic estimated-weight correction, not ordinary ML/FIML robust-score normal-theory inference. The complete-data WLS/ADF empirical-weight slice has also landed, using (3) for the full dense $\dot{\Gamma}_i$ and a finite-difference contamination gate for both covariance-only and meanstructure layouts. Complete-data DWLS has landed as the diagonal extraction of the same empirical-Gamma influence, gated by case-weight finite differences of $\text{diag}(\hat{\Gamma})^{-1}$ in covariance-only and meanstructure layouts. Complete-data DLS with fixed mixing scalar has also landed as the convex mixture of the sample-built normal-theory and dense empirical-Gamma

influence channels, with finite-difference gates on the mixed inverse weight in covariance-only and meanstructure layouts. The raw mean/covariance g_i is analytic; EB-DLS scalar-selection uncertainty remains separate from the fixed- a weight derivative.

Two-stage FIML (ML2S). Stage 1 supplies saturated EM moments and their ACOV. The observed-bread single-model regime has landed as **TwoStageBread::Observed**: it changes the Stage-2 bread while leaving the Stage-1 ACOV and the current fixed-weight robust sandwich unchanged. Complete-data gates reduce the NT path to unstructured observed-bread robust SEM and the ADF path to observed-bread **robust_continuous_ls**. The casewise saturated-moment influence under the missingness pattern has also landed as **saturated_em_moment_influence**. Let $s_i(\hat{\eta}_S) = \partial \ell_i(\eta)/\partial \eta|_{\hat{\eta}_S}$ be the observed-pattern saturated log-likelihood score for $\eta = (\mu, \text{vech } \Sigma)$, and let $I_\eta = -\sum_i \partial^2 \ell_i(\eta)/\partial \eta \partial \eta^\top|_{\hat{\eta}_S}$. The exported row is

$$a_i^\top = s_i(\hat{\eta}_S)^\top I_\eta^{-1}, \quad A^\top A = I_\eta^{-1} \left(\sum_i s_i s_i^\top \right) I_\eta^{-1} = \widehat{\text{ACOV}}(\hat{\eta}_S). \quad (4)$$

For complete data this reduces blockwise to

$$a_i = n_b^{-1} [x_i - \bar{x}_b, \text{vech}\{(x_i - \bar{x}_b)(x_i - \bar{x}_b)^\top - \hat{\Sigma}_b\}].$$

For complete raw data, the raw-data **two_stage_em_ml_inference** overload now delegates observed-bread **Dwls**, **Adf**, and fixed- a **Dls** covariance to the already-gated continuous-LS IJ adapters; this is an exact reduction because saturated EM moments reduce to ordinary mean/covariance moments and $n_b A_b^\top A_b$ reduces to Browne’s complete data $\hat{\Gamma}$. The scaled-test fields remain the fixed-weight Satorra–Bentler quantities because Hall–Inoue test-statistic corrections are estimator-specific and have not been derived here.

For missing data, the remaining ML2S IJ adapter is not just to reuse a_i . The **Dwls**, **Adf**, and **Dls** weights depend on $\Gamma_{\text{FIML}} = n_b I_\eta^{-1} (\sum_i s_i s_i^\top) I_\eta^{-1}$, so their weight derivative needs the Stage-1 sandwich-Gamma influence: the score-product channel, the observed-information channel, and the movement of (I_η, J_η) through $\hat{\eta}_S$. The default **TwoStageWeight::Nt** route in **two_stage_em_ml_inference** is the normal-theory ML robust-score path; the moment-quadratic GLS normal-theory-weight IJ correction remains available as **robust_continuous_ls_gls_ij** for complete continuous LS rather than being silently substituted for ML2S NT.

MCAR and pairwise-missing ordinal/polyserial. The current primitive assumes disjoint blocks: it sums $V_b^\top V_b$ inside each block. Pairwise missingness has overlapping supports. The correct object is one case-aligned sparse row vector over the full moment stack, with zeros for unobserved moment components and the right N/n_{jk} scaling for each pair-local estimator. This probably needs a second primitive. An extension of the current IJ block would need case ids and per-moment support scaling. The gates are complete-data reduction, MCAR simulation against a case-resampling jackknife, and equality of pairwise-overlap covariance to the existing NACOV.

6 Verification standard

Every adapter gets the same four checks before it is trusted:

1. fixed-weight reduction to `robust_weighted_moments`;
2. finite-difference or case-weight derivative check for each new $\dot{\Gamma}_i$ channel;
3. resampling-jackknife agreement on a small deterministic fixture;
4. simulation calibration under the misspecified model that motivated the method, with ULS/fixed-weight cells used as the negative control.

References

- [1] Hall, A. R., and Inoue, A. (2003). The large sample behaviour of the generalized method of moments estimator in misspecified models. *Journal of Econometrics*, 114(2), 361–394. doi:10.1016/S0304-4076(03)00075-0.